B. Brightness Begins

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: binary search, math | https://codeforces.com/problemset/problem/2020/B

Imagine you have n light bulbs numbered 1, 2, …, n. **Initially, all bulbs are on**. To *flip* the state of a bulb means to turn it off if it used to be on, and to turn it on otherwise.

Next, you do the following:

- for each $i = 1, 2, \ldots, n$, flip the state of all bulbs j such that j is divisible by i^{dagger}.

After performing all operations, there will be several bulbs that are still on. Your goal is to make this number exactly k.

Find the smallest suitable n such that after performing the operations there will be exactly k bulbs on. We can show that an answer always exists.

dagger An integer x is divisible by y if there exists an integer z such that $x = y \cdot z$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The only line of each test case contains a single integer k ($1 \le k \le 10^{18}$).

Output

For each test case, output n — the minimum number of bulbs.

Example

Input
3 1 3 8

Output
2 5 11

Note

In the first test case, the minimum number of bulbs is 2. Let's denote the state of all bulbs with an array, where 1 corresponds to a turned on bulb, and 0 corresponds to a turned off bulb. Initially, the array is [1, 1].

- After performing the operation with $i = 1$, the array becomes [underline0, underline0].
- After performing the operation with $i = 2$, the array becomes [0, underline1].

In the end, there are $k = 1$ bulbs on. We can also show that the answer cannot be less than 2.

In the second test case, the minimum number of bulbs is 5. Initially, the array is [1, 1, 1, 1, 1].

- After performing the operation with $i = 1$, the array becomes [underline0, underline0, underline0, underline0, underline0].
- After performing the operation with $i = 2$, the array becomes [0, underline1, 0, underline1, 0].
- After performing the operation with $i = 3$, the array becomes [0, 1, underline1, 1, 0].
- After performing the operation with $i = 4$, the array becomes [0, 1, 1, underline0, 0].
- After performing the operation with $i = 5$, the array becomes [0, 1, 1, 0, underline1].

In the end, there are $k = 3$ bulbs on. We can also show that the answer cannot be smaller than 5.